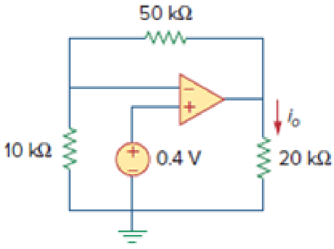
Problem

Find *io* in the op amp circuit of Fig. 5.66.

Figure. 5.66:



Step-by-step solution

Step 1 of 4

Given circuit is,

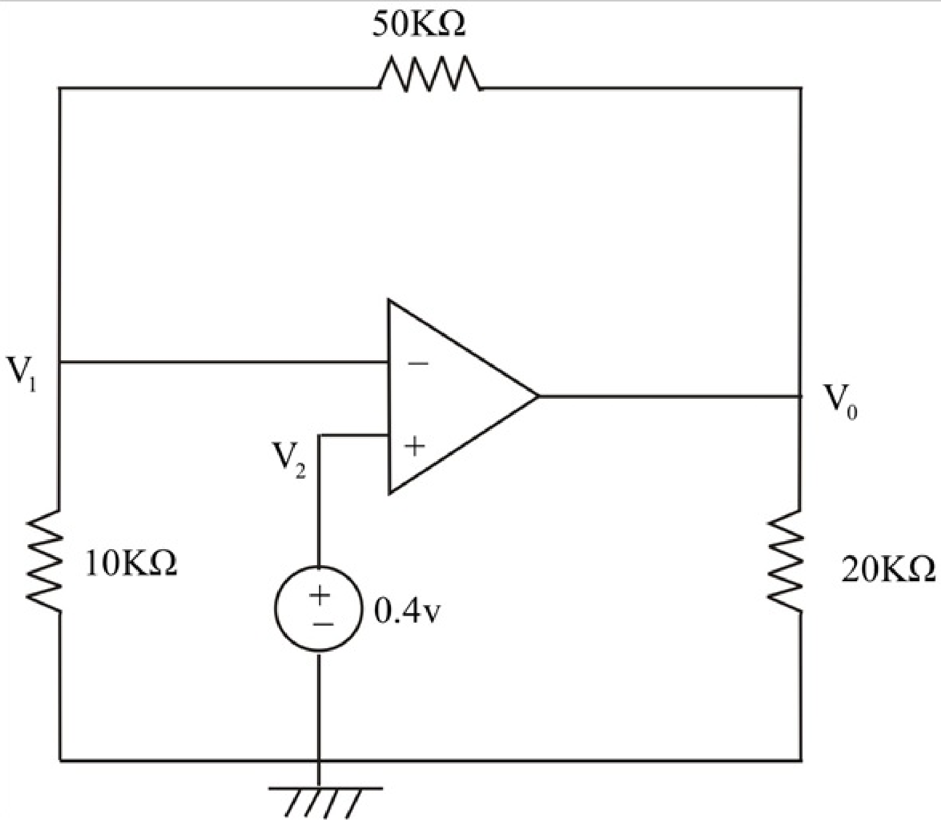


Fig-1

Step 2 of 4

From Fig-1,

At node 1,

$$\frac{0-V_1}{10\times10^3}=\frac{V_1-V_0}{50\times10^3} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

But,

$$V_1 = 0.4\text{V}$$

From Equation-(1),

$$-5V_1 = V_1 - V_0$$

$$V_0 = 6V_1$$

$$V_0 = 6\times0.4$$

$$V_0 = 2.4\text{V}$$

Step 3 of 4

Alternatively, viewed as a non-inverting amplifier,

For non-inverting amplifier gain is,

$$V_0 = \left(1+\frac{R_f}{R_i}\right)V_i$$

$$V_0 = \left(1+\left(\frac{50}{10}\right)\right)\times0.4$$

Therefore,

$$V_0 = 2.4\text{V}$$

Step 4 of 4

Therefore,

$$i_0 = \frac{V_0}{R_{20\text{k}\Omega}}$$

$$i_0 = \frac{2.4}{20\times10^3}$$

$$\boxed{i_0 = 0.12\text{mA}}$$